SBA LOAN ANALYSIS

Capstone Project Report | Python • MySQL • Power BI

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Project Component	Technology / Output
Data cleaning and exploratory analysis	Python / Jupyter Notebook
Relational storage and SQL analysis	MySQL
Interactive reporting and visualization	Power BI
Dataset size	5,000 records × 18 columns

Abstract

This project analyzes a sample of 5,000 Small Business Administration (SBA) loan records using Python, MySQL, and Power BI. The workflow covers data inspection, cleaning, exploratory summaries, SQL-based business analysis, and an interactive dashboard. The analysis focuses on loan approval amounts, fiscal-year trends, loan status, charge-off amounts, interest rates, loan terms, business age, processing methods, and jobs supported.

The cleaned dataset contains 18 columns and was loaded into a MySQL database for structured analysis. Eight SQL views were created to make common summaries reusable, and additional SQL work demonstrated aggregation, an INNER JOIN, and a window function ranking states by total approval amount.

1. Introduction

Small-business lending data can help analysts understand lending volume, the distribution of loan statuses, geographic concentration, and recorded charge-offs. This capstone project demonstrates an end-to-end workflow from data preparation through SQL analysis to business intelligence reporting.

1.1 Problem Statement

Loan-level datasets may contain inconsistent data types, missing fields, and many records that are difficult to interpret directly. The project addresses the need to prepare records consistently and provide filterable views of lending activity.

1.2 Objectives

- Inspect the dataset and identify missing values, duplicates, and invalid numeric values.
- Clean key fields and export a reusable processed dataset.
- Load the records into MySQL and answer business questions with SQL.
- Create reusable summary views and demonstrate JOIN and window-function analysis.
- Build a multi-page Power BI dashboard with slicers, KPIs, and charts.
- Summarize findings and provide practical recommendations for additional analysis.

1.3 Scope

The project covers 5,000 records spanning fiscal years 2020–2026 in the supplied dataset. It examines approval values, guaranteed approval, interest rates, terms, business classifications, state, status, charge-off amounts, and jobs supported.

2. Tools and Technologies

Tool	Role in the Project
Python / pandas	Read, inspect, clean, transform, and summarize data.
Jupyter Notebook	Document and execute the Python workflow.
MySQL	Store the cleaned records and run repeatable analytical SQL.
Power BI	Create interactive report pages, KPIs, slicers, and visual summaries.

2.1 Workflow

1. Load the source CSV and inspect dimensions, columns, data types, and quality.
2. Handle missing business classification values and parse date fields.
3. Validate numeric fields and export the cleaned dataset.
4. Load cleaned records into the MySQL table `sba_loan_data`.
5. Run business-question queries and create reusable views.
6. Connect Power BI to the dataset and build three dashboard pages.

3. Dataset Description

The dataset contains 5,000 rows and 18 columns. The fields describe approval dates and fiscal years, processing methods, gross and guaranteed approval amounts, interest rates, loan terms, industry descriptions, project state, business type and age, loan status, paid-in-full and charge-off dates, charge-off amount, and jobs supported.

Field Group	Representative Columns	Analytical Use
Time	ApprovalDate, ApprovalFY	Trend analysis by date or fiscal year.
Loan amount	GrossApproval, SBAGuaranteedApproval	Funding and approval summaries.
Loan characteristics	InitialInterestRate, TermInMonths, ProcessingMethod	Rate, term, and program comparisons.
Business/geography	NaicsDescription, ProjectState, BusinessType, BusinessAge	Industry, state, and segment summaries.
Outcome / impact	LoanStatus, GrossChargeOffAmount, JobsSupported	Status distribution, charge-offs, supported jobs.

3.1 Initial Data Quality Review

Data Quality Check	Before Cleaning	After Cleaning
Missing BusinessType	1	0
Missing BusinessAge	2	0
Missing PaidInFullDate	4,122	4,122
Missing ChargeOffDate	4,911	4,911
Duplicate rows	0	0
Total missing values	9,036	9,033
Dataset size	5,000 × 18	5,000 × 18

The missing paid-in-full and charge-off dates are plausibly related to loan outcomes rather than universally invalid values: 878 records have a paid-in-full date and 89 have a charge-off date.

5. MySQL Database and SQL Analysis

The MySQL database is named `sba_loan_analysis`, and the main table is `sba_loan_data` containing 5,000 records and 18 columns.

5.1 Star Schema Design

The project uses a star schema design with one central fact table (`fact_sba_loan`) supported by four dimension tables: `dim_approval_date`, `dim_state`, `dim_business`, and `dim_loan_details`.

5.2 Fact Table Structure and Validation

The `fact_sba_loan` table stores loan-level records and numerical measures used for analytical reporting.

Column	Purpose
LoanID	Primary key that uniquely identifies each loan record
DateKey	Foreign key linking to dim_approval_date
StateKey	Foreign key linking to dim_state
BusinessKey	Foreign key linking to dim_business
LoanDetailsKey	Foreign key linking to dim_loan_details
GrossApproval	Gross loan approval amount
SBAGuaranteedApproval	SBA-guaranteed approval amount
InitialInterestRate	Initial interest rate
TermInMonths	Loan term in months
GrossChargeOffAmount	Recorded gross charge-off amount
JobsSupported	Number of jobs supported

Validation results:

- Total records in the fact table: 5,000.
- LoanID is defined as the primary key.
- Four foreign-key relationships connect the fact table to the dimension tables.
- All four dimension relationships were validated.
- Missing dimension-key matches: 0.
- The fact table supports analysis of loan approvals, charge-offs, interest rates, loan terms, and jobs supported across the dimension categories.

5.3 Summary Views

View Name	Purpose
vw_loan_status_summary	Summarize records by loan status.
vw_yearly_loan_summary	Summarize lending metrics by fiscal year.
vw_processing_method_summary	Compare processing methods.
vw_state_loan_summary	Summarize loan activity by state.
vw_business_age_summary	Compare business-age categories.
vw_loan_term_summary	Summarize loan terms.
vw_jobs_supported_summary	Summarize job-support metrics.
vw_interest_rate_summary	Summarize interest-rate metrics.

5.4 State Summary and JOIN

A state_loan_summary aggregate table was created with total loans, total approval amount, and total charge-off amount grouped by ProjectState. An INNER JOIN between the loan-level table and this summary table was then used to demonstrate joining data sources. Grouping the result by state and the summary columns collapses repeated loan-level matches to one output row per state.

5.5 Window Function

A `RANK() OVER (ORDER BY TotalApprovalAmount DESC)` expression ranked states by total approval amount. In the resulting 54-state/project-state-category rows, California ranked first, Texas second, and Florida third. This is a ranking of aggregate approval dollars, not a ranking of loan quality or performance.

5.6 Example Business Questions

- How many loan records appear in each loan status?
- How do total approval amounts change by fiscal year?
- Which states have the largest total approval amounts?
- How do processing methods compare on volume and recorded charge-offs?
- How do business-age categories compare on recorded charge-off counts?
- What are the average interest rate and loan term?
- How many jobs are supported across fiscal years?
- Which states rank highest by total approval amount?
- How is recorded charge-off amount distributed by status?
- How do loan volumes vary by business type and geography?
- How do summary metrics compare across loan term groups?

5.7 Data Warehouse Design – Star Schema

To improve analytical organization and support efficient Power BI reporting, a star schema was designed in MySQL. The schema contains one central fact table and four dimension tables.

Fact Table:

- `fact_sba_loan` – Stores 5,000 loan records and measurable values such as gross approval, guaranteed approval, charge-off amount, jobs supported, and foreign keys linking to the dimensions.

Dimension Tables:

Dimension Table	Records	Purpose
dim_state	54	Stores state information
dim_approval_date	1,604	Stores approval date and fiscal-year information
dim_business	1,649	Stores business classification and age details
dim_loan_details	71	Stores loan characteristics and status details

Relationships:
The fact table connects to all four dimension tables through their respective keys:

- StateKey → dim_state
- DateKey → dim_approval_date
- BusinessKey → dim_business
- LoanDetailsKey → dim_loan_details

Each relationship follows a one-to-many structure from the dimension table to the fact table. Foreign-key validation confirmed that all 5,000 fact records have matching dimension records.

This design separates measurable loan transactions from descriptive attributes. It supports filtering, grouping, and comparing loan performance across time, geography, business categories, and loan statuses in Power BI.

6. Exploratory Analysis and Results

6.1 Overall Metrics

Metric	Observed Result
Records	5,000
Total gross approval	Approximately $2.57 billion
Average gross approval per record	$513,860.90
Median gross approval	$198,000
Average initial interest rate	8.86%
Average loan term	138.05 months
Total recorded gross charge-off amount	$10,874,334.62
Records with CHGOFF status	89 (1.78% of records)
Jobs supported	Approximately 52,000

6.2 Loan Status Distribution

Loan Status	Records	Share of 5,000 Records
EXEMPT	3,119	62.38%
P I F	878	17.56%
CANCLD	642	12.84%
COMMIT	272	5.44%
CHGOFF	89	1.78%

Interpretation of EXEMPT Loan Status: According to the SBA 7(a) and 504 Loan Data Dictionary, EXEMPT refers to loans that have been disbursed but have not been cancelled, paid in full, or charged off. These loans are exempt from public disclosure under FOIA Exemption 4.

6.3 Fiscal-Year Trend

Fiscal Year	Loans	Total Approval	Jobs Supported
2020	550	$287,757,400	5,884
2021	678	$448,488,100	7,679
2022	563	$313,674,100	6,423
2023	713	$347,253,900	7,854
2024	913	$411,209,700	8,558
2025	1,071	$515,800,800	10,532
2026	512	$245,120,500	5,135

2025 has the highest loan count and total approval among the fiscal years listed. The 2026 total is lower, but the period may be incomplete; it should not be treated as a full-year decline without checking the dataset's coverage date.

6.4 Processing Method Comparison

Processing Method	Loans	CHGOFF Records	CHGOFF Share
Preferred Lenders Program	2,750	29	1.05%
SBA Express Program	1,888	53	2.81%
Community Advantage Initiative	41	3	7.32%

In this sample, the SBA Express Program has a higher observed CHGOFF record share than the Preferred Lenders Program. The Community Advantage Initiative has only 41 records, so its 7.32% share is unstable and should be interpreted cautiously. These are descriptive comparisons and do not establish that the processing method caused the observed differences.

6.5 Business-Age Comparison

Business Age Category	Loans	CHGOFF Records	CHGOFF Share
Existing or more than 2 years old	2,847	49	1.72%
Startup, Loan Funds will Open Business	801	17	2.12%
New Business or 2 years or less	778	14	1.80%
Change of Ownership	522	6	1.15%
Unanswered	50	2	4.00%

Business Age Category	Loans	CHGOFF Records	CHGOFF Share
Unknown	2	1	50.00%

The Unknown category has only two records and should not be used for a meaningful risk conclusion. The Unanswered group also has a small sample. The largest group by volume is existing businesses, and it also accounts for the greatest total charge-off dollars due to its larger record count.

7. Power BI Dashboard

The Power BI report contains three pages with slicers, KPI cards, and charts. The dashboard is designed to let viewers filter the sample by year, state, loan status, and business type and then explore different aspects of the lending data.

7.1 Executive Summary

- **Slicers:** Year, Project State, Loan Status, Business Type.
- **KPI cards:** Total Loan Approval, Total Loans, Total Charge-offs, Jobs Supported, Average Interest Rate, Average Loan Approval.
- **Charts:** Loan Approval Trend by Fiscal Year, Loan Status Distribution, Charge-offs by Fiscal Year.

7.2 Loan Risk & Status Analysis

- **Slicers:** ApprovalFY, Project State, BusinessType, Loan Status.
- **KPI cards:** Total Loan Approval, Total Charge-Off, Average Interest Rate.
- **Charts:** Charge-off by Loan Status, Average Interest Rate by Business Type, Number of Loans by Status.

7.3 Business & Economic Impact

- **KPI cards:** Total Jobs Supported, Total Loan Approval, Average Loan Term.
- BusinessAgeShort classification groups raw business-age labels into Existing Business, Startup, New Business, Ownership Change, Unanswered, and Unknown.
- Average loan term is formatted to zero decimal places for readability.

The dashboard uses a clean white background and violet chart styling. Before final submission, check all slicers interact with the intended visuals, confirm that totals match SQL summaries under the same filters, and ensure percentage measures use the correct units.

The four active relationships between the fact table and dimension tables allow slicers to filter the star-schema-based measures. Slicer interactions were tested across all three dashboard pages to confirm that the KPI cards and charts respond to the selected filters.

7.4 DAX Measures and Validation

Power BI DAX measures were created to calculate key loan performance indicators and support interactive analysis. A dedicated Measure Table was used to organize the original dataset measures and star schema measures separately from the data tables.

Key DAX Measures:

Measure	Purpose
Total Loan Approval Star	Calculates the total gross approval amount from the fact table.
Total Loans Star	Counts distinct loan records using LoanID.
Total Charge-offs Star	Calculates the total recorded gross charge-off amount.
Jobs Supported Star	Calculates the total number of jobs supported.
Average Interest Rate Star	Calculates the average initial interest rate.
Average Loan Term Star	Calculates the average loan term in months.
Charge-off Rate Star	Calculates the percentage of records with CHGOFF status.
Previous Year Approval Star	Calculates the previous fiscal-year approval amount.
Approval Growth % Star	Calculates the percentage change in approval amount compared with the previous year.

Validation Results

The original dataset measures were compared with the corresponding star schema measures to verify consistency.

Metric	Original Dataset	Star Schema
Total Loan Approval	$2.569 billion	$2.569 billion
Total Loans	5,000	5,000
Total Charge-offs	$10.874 million	$10.874 million

The comparison showed matching results for these key metrics, confirming consistency between the original dataset and the star schema calculations.

The average initial interest rate was validated at approximately 8.86%. The measure was formatted as a percentage with two decimal places, ensuring that the displayed value matched the source data's percentage-point representation.

The charge-off rate was calculated as approximately 1.78%, based on 89 CHGOFF records out of 5,000 total loan records.

The dashboard slicers were also tested to confirm that the KPI cards and charts respond to selections for fiscal year, state, loan status, business type, and business age, as applicable to each report page.

These validation checks help ensure that the Power BI dashboard provides consistent and interactive summaries of the loan dataset.

KPI Validation

The key Power BI dashboard measures were validated against the SQL analysis results to ensure consistency between the database and dashboard. Total Loan Approval was approximately $2.569 billion, Total Loans = 5,000, and Total Charge-offs were approximately $10.874 million. The corresponding Power BI measures produced matching results, confirming that the dashboard KPIs accurately represent the underlying loan dataset.

8. Key Findings and Recommendations

8.1 Key Findings

- The cleaned analysis dataset contains 5,000 records and 18 columns, with no duplicate rows detected.
- 2025 is the largest listed fiscal year by loan count (1,071) and total approval ($515.8 million).
- California, Texas, and Florida rank as the top three states by total approval amount in the state-ranking output.
- The EXEMPT status is the largest category (62.38%); its interpretation requires the official data dictionary.
- There are 89 CHGOFF-status records (1.78% of records) and $10.87 million in recorded gross charge-off amount.
- SBA Express has a higher observed CHGOFF record share than the Preferred Lenders Program in this sample; this is descriptive, not causal.
- The Community Advantage and Unknown business-age groups have small counts, so their percentages are sensitive to individual records.

8.2 Recommendations

- Confirm status codes, especially EXEMPT, against the source data dictionary before using them as outcome labels.
- Use complete fiscal-year coverage when comparing trends; flag 2026 as potentially partial.
- For risk comparisons, report both counts and rates, and include sample sizes to avoid overinterpreting small groups.
- Improve the MySQL schema by storing date columns as DATE/DATETIME and adding suitable keys and indexes.
- Validate Power BI KPI totals against SQL outputs and document any filters or business rules.
- For deeper risk analysis, consider loan-level controls such as loan amount, term, business type, industry, and geography; avoid causal claims from simple group comparisons.
- Add a README containing setup instructions, data dictionary notes, query examples, dashboard screenshots, and known limitations.

9. Limitations

- The analysis is based on the supplied 5,000-record extract and may not represent all SBA lending activity.
- The 2026 fiscal-year records may be incomplete.
- The meaning of some status values, especially EXEMPT, requires confirmation from official documentation.
- Date fields are stored as TEXT in the current MySQL schema, limiting date-specific SQL operations.
- Observed group differences do not prove causation and may be influenced by differences in loan composition.
- Small categories produce unstable percentages; the sample size should always be shown alongside the rate.
- Dashboard totals should be reconciled with SQL results under identical filters before final presentation.

10. Conclusion

This capstone demonstrates a practical analytics pipeline for SBA loan records using Python, MySQL, and Power BI. Python was used to inspect and prepare the data, MySQL supported reusable summaries and business-question queries, and Power BI delivered interactive views of lending volume, status, charge-offs, and economic impact. The analysis highlights changes in annual loan activity, geographic concentration, and differences across processing methods and business-age categories. The results provide a foundation for further investigation, while data definitions, period coverage, small sample sizes, and consistency checks remain important considerations.

Appendix A: Technical Inventory

Artifact	Description
sample.ipynb	Python notebook for loading, cleaning, and exploring data.
data/SBALoanData-LoanApproval_DefaultAnalysis.csv	Raw source CSV.
data/processed/sba_loan_cleaned.csv	Cleaned dataset (5,000 × 18).
sba_loan_analysis.sba_loan_data	Main MySQL table.
Eight vw_* views	Reusable SQL summary views.
state_loan_summary	Aggregate state table used for JOIN and RANK demonstration.
Power BI PBIX	Three-page interactive dashboard.

Appendix B: Reproducibility Checklist

- Keep the raw CSV unchanged and document its original source.
- Run the notebook from top to bottom and verify the cleaned row/column count.
- Load the cleaned CSV into MySQL and verify 5,000 records.
- Run the SQL views and ranking query and compare totals.
- Refresh the Power BI model and test slicer interactions.